Write CREATE TABLE statements for all 5 tables with correct data types, PRIMARY KEY and FOREIGN KEY constraints.

```
CREATE TABLE customers (
    customer_id VARCHAR(20) PRIMARY KEY,
    customer_name VARCHAR(100) ,
    email VARCHAR(100),
    phone NUMERIC,
    age NUMERIC CHECK (age >= 0),
    gender VARCHAR(20),
    city VARCHAR(20),
    state VARCHAR(20),
    registration_date DATE ,
    loyalty_tier VARCHAR(20) ,
    is_active VARCHAR(10)
);
CREATE TABLE marketing (
    campaign_id VARCHAR(20) PRIMARY KEY,
    campaign_name VARCHAR(30) ,
    channel VARCHAR(30),
    start_date DATE ,
    duration_days NUMERIC,
    spend_inr NUMERIC,
    impressions NUMERIC ,
    clicks NUMERIC,
    conversions NUMERIC ,
    revenue_generated NUMERIC,
    ab_group VARCHAR(20),
    target_segment VARCHAR(20)
);
CREATE TABLE orders (
    order_id VARCHAR(30) PRIMARY KEY,
    customer_id VARCHAR(20) REFERENCES customers(customer_id),
    product_id VARCHAR(20),
    order_date DATE ,
    quantity NUMERIC,
    unit_price NUMERIC,
    discount_pct NUMERIC,
```

```
    discount_amount NUMERIC(10, 2),
    total_amount NUMERIC(12, 2),
    payment_method VARCHAR(20),
    order_status VARCHAR(20),
    delivery_days NUMERIC,
    region VARCHAR(10)
);
CREATE TABLE products (
    product_id VARCHAR(20) PRIMARY KEY,
    product_name VARCHAR(30),
    category VARCHAR(30),
    cost_price NUMERIC,
    selling_price NUMERIC,
    margin_pct NUMERIC(5, 2),
    stock_qty NUMERIC,
    supplier VARCHAR(50),
    rating NUMERIC(3, 2)
);
CREATE TABLE returns (
    return_id VARCHAR(30) PRIMARY KEY,
    order_id VARCHAR(30) REFERENCES orders(order_id),
    return_date DATE,
    return_reason VARCHAR(50),
    refund_amount NUMERIC(12, 2),
    refund_status VARCHAR(20)
);
```

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Select all orders placed in 2024. Show order_id, customer_id, order_date, total_amount.
SELECT order_id, customer_id, order_date, total_amount
FROM orders
WHERE order_date BETWEEN '2024-01-01' AND '2024-12-31'
ORDER BY order_date ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 3

	order_id [PK] character varying (30)	customer_id character varying (20)	order_date date	total_amount numeric (12,2)
1	OD-20240101-ZT4A7SDT	RXC-3074AAD4	2024-01-01	50997.00
2	OD-20240101-E6TWCVZ2	RXC-F1B38963	2024-01-01	10636.20
3	OD-20240101-QVVA34P5	RXC-B68FCD93	2024-01-01	899.10
4	OD-20240101-3QER0SJJ	RXC-FC8CA9DA	2024-01-01	853.10
5	OD-20240102-H6C8E6PT	RXC-FBB201B0	2024-01-02	521.55
6	OD-20240102-SHV65R87	RXC-196A9B81	2024-01-02	1010.25
7	OD-20240102-80SSNJWV	RXC-7C4CE1B7	2024-01-02	1274.15
8	OD-20240103-JJARQN4X	RXC-6D4E2245	2024-01-03	4936.20
9	OD-20240103-EB77ZG61	RXC-B118FA82	2024-01-03	424.65
10	OD-20240103-PV501JWX	RXC-5784E1DC	2024-01-03	551.65
11	OD-20240103-8JE745G4	RXC-2993B681	2024-01-03	1899.05
12	OD-20240103-3CWTG57W	RXC-4AA96371	2024-01-03	616.55
13	OD-20240103-VBBY26QV	RXC-CF36AA15	2024-01-03	948.10
14	OD-20240103-GA1N86QF	RXC-70E423D5	2024-01-03	11899.15
15	OD-20240104-BV7PXJ7Y	RXC-E9A17CE7	2024-01-04	1168.20
16	OD-20240104-CZ86E4RX	RXC-415241C2	2024-01-04	149.00
17	OD-20240105-AH5GFMVK	RXC-60A5B343	2024-01-05	1499.00
18	OD-20240105-F7MZPR4C	RXC-1259C158	2024-01-05	2997.00

Total rows: 2711 | Query complete 00:00:00.098 | CRLF | Ln 6, Col 1

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```
7   -- List distinct payment methods and count of orders for each. Sort by count descending.
8   SELECT payment_method, COUNT(order_id) AS order_count
9   FROM orders
10  GROUP BY payment_method
11  ORDER BY order_count DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 6 | Page No: 1 of 1

	payment_method character varying (20)	order_count bigint
1	UPI	3021
2	Credit Card	1902
3	Debit Card	1173
4	Cash on Delivery	769
5	Net Banking	750
6	Wallet	385

Successfully run. Total query runtime: 125 msec. 6 rows affected.

Total rows: 6 | Query complete 00:00:00.125 | CRLF | Ln 11, Col 27

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Find all orders where discount_pct > 10 AND order_status = 'Delivered'. Use aliases for column names.
SELECT order_id AS "Order ID",
    customer_id AS "Customer ID",
    order_date AS "Date of Order",
    discount_pct AS "Discount Percentage",
    discount_amount AS "Discount Amount (INR)",
    total_amount AS "Final Invoice Amount",
    order_status AS "Status"
FROM orders
WHERE discount_pct > 10
  AND order_status = 'Delivered'
ORDER BY order_date DESC;
```

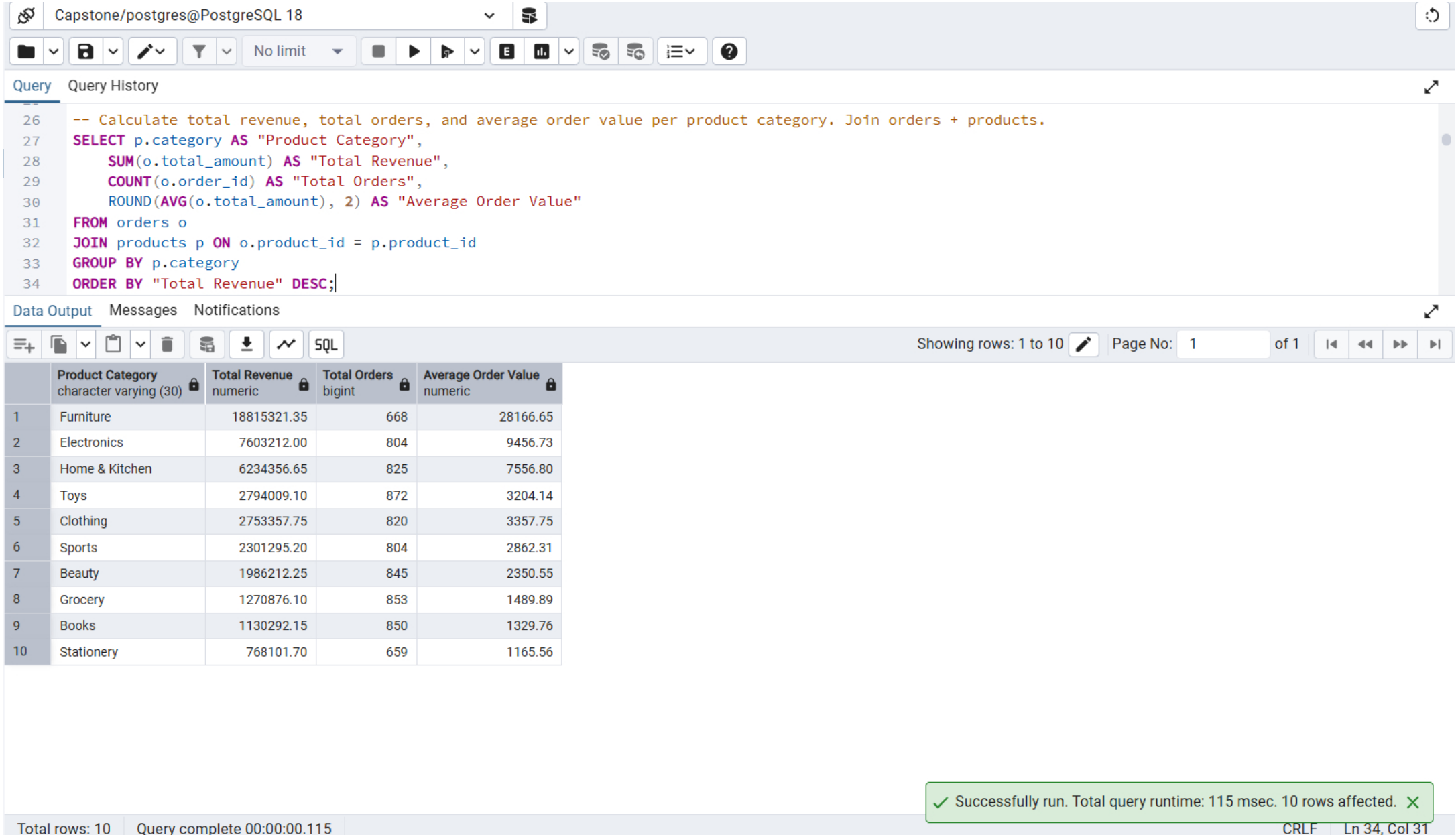
Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 2

	Order ID character varying (30)	Customer ID character varying (20)	Date of Order date	Discount Percentage numeric	Discount Amount (INR) numeric (10,2)	Final Invoice Amount numeric (12,2)	Status character varying (20)
1	OD-20241229-MVRNJA...	RXC-7D0B5C70	2024-12-29	15	224.70	1273.30	Delivered
2	OD-20241228-03ZHWKP...	RXC-503F6835	2024-12-28	15	149.85	849.15	Delivered
3	OD-20241228-8MDBY10Z	RXC-DCA54184	2024-12-28	25	1948.50	5845.50	Delivered
4	OD-20241227-WP3H2M...	RXC-738B6B94	2024-12-27	20	199.60	798.40	Delivered
5	OD-20241227-JDA7NST4	RXC-30170216	2024-12-27	15	5099.70	28898.30	Delivered
6	OD-20241226-G7STGMK3	RXC-57F847B9	2024-12-26	25	249.50	748.50	Delivered
7	OD-20241226-BM366PRN	RXC-FE83F7CF	2024-12-26	15	254.70	1443.30	Delivered
8	OD-20241226-QFSWJS3R	RXC-F2A5E31D	2024-12-26	15	599.85	3399.15	Delivered
9	OD-20241226-ECQXCN1B	RXC-9E48CC05	2024-12-26	20	659.40	2637.60	Delivered
10	OD-20241225-43PJBBR4	RXC-F84A476A	2024-12-25	20	2879.40	11517.60	Delivered
11	OD-20241225-E2DM1Q...	RXC-CC7A402C	2024-12-25	25	249.75	749.25	Delivered
12	OD-20241223-C5X3GHPE	RXC-3A5C4B60	2024-12-23	15	194.70	1103.30	Delivered
13	OD-20241223-NHK4Q44H	RXC-DFC38B2F	2024-12-23	15	89.70	508	

Successfully run. Total query runtime: 124 msec. 1590 rows affected.

Total rows: 1590 | Query complete 00:00:00.124 | CRLF | Ln 23, Col 33



	Product Category character varying (30)	Total Revenue numeric	Total Orders bigint	Average Order Value numeric
1	Furniture	18815321.35	668	28166.65
2	Electronics	7603212.00	804	9456.73
3	Home & Kitchen	6234356.65	825	7556.80
4	Toys	2794009.10	872	3204.14
5	Clothing	2753357.75	820	3357.75
6	Sports	2301295.20	804	2862.31
7	Beauty	1986212.25	845	2350.55
8	Grocery	1270876.10	853	1489.89
9	Books	1130292.15	850	1329.76
10	Stationery	768101.70	659	1165.56

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Find the month-wise revenue trend for 2023 and 2024. Use date functions to extract year and month.
SELECT EXTRACT(YEAR FROM order_date) AS order_year,
       EXTRACT(MONTH FROM order_date) AS order_month,
       SUM(total_amount) AS total_revenue,
       COUNT(order_id) AS total_orders
FROM orders
WHERE EXTRACT(YEAR FROM order_date) IN (2023, 2024)
GROUP BY EXTRACT(YEAR FROM order_date), EXTRACT(MONTH FROM order_date)
ORDER BY order_year ASC, order_month ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 24 | Page No: 1 of 1

	order_year numeric	order_month numeric	total_revenue numeric	total_orders bigint
1	2023	1	805785.15	170
2	2023	2	769443.35	135
3	2023	3	1208518.05	195
4	2023	4	1089535.60	211
5	2023	5	940674.50	180
6	2023	6	1245682.05	187
7	2023	7	1040931.45	185
8	2023	8	788190.30	182
9	2023	9	1189134.95	250
10	2023	10	1570104.35	346
11	2023	11	2056015.55	349
12	2023	12	1494833.50	265
13	2024	1	1180842.80	175
14	2024	2	632030.30	148
15	2024	3	938849.20	188
16	2024	4	1005036.15	192

Successfully run. Total query runtime: 75 msec. 24 rows affected.

Total rows: 24 | Query complete 00:00:00.075 | CRLF | Ln 44, Col 42

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Identify customers who have placed more than 5 orders. List name, tier, and order count.
SELECT c.customer_name AS "Customer Name",
    c.loyalty_tier AS "Loyalty Tier",
    COUNT(o.order_id) AS "Order Count"
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.customer_name, c.loyalty_tier
HAVING COUNT(o.order_id) > 5
ORDER BY "Order Count" DESC, "Customer Name" ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 440 | Page No: 1 of 1

	Customer Name character varying (100)	Loyalty Tier character varying (20)	Order Count bigint
1	Karan Iyer	Silver	11
2	Rohit Chopra	Bronze	11
3	Suresh Kumar	Bronze	11
4	Tarun Dubey	Gold	11
5	Vivek Joshi	Silver	11
6	Deepa Agarwal	Bronze	10
7	Gaurav Kumar	Bronze	10
8	Kavya Malhotra	Bronze	10
9	Lakshmi Malhotra	Bronze	10
10	Naveen Joshi	Bronze	10
11	Neha Verma	Bronze	10
12	Pooja Srivastava	Gold	10
13	Ritika Chopra	Silver	10
14	Sneha Chaudhary	Platinum	10
15	Aarav Dubey	Bronze	9
16	Amit Sharma	Bronze	9

Successfully run. Total query runtime: 132 msec. 440 rows affected.

Total rows: 440 | Query complete 00:00:00.132 | CRLF | Ln 54, Col 50

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Write a CASE statement to label orders: 'High Value' if total_amount > 5000, 'Medium' if > 1500, else 'Low'.
SELECT order_id, customer_id, total_amount,
    CASE
        WHEN total_amount > 5000 THEN 'High Value'
        WHEN total_amount > 1500 THEN 'Medium'
        ELSE 'Low'
    END AS value_band
FROM orders
ORDER BY total_amount DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 8

	order_id [PK] character varying (30)	customer_id character varying (20)	total_amount numeric (12,2)	value_band text
1	OD-20220821-VFXPPSBQ	RXC-64AE43C9	314991.00	High Value
2	OD-20221106-1ZS8XDKH	RXC-5E5BC156	251992.80	High Value
3	OD-20220424-800MRHNE	RXC-6A24D66F	251992.80	High Value
4	OD-20220716-K7VQSTBE	RXC-0F2D15A2	209994.00	High Value
5	OD-20221125-XPDV9DFX	RXC-661BC156	209994.00	High Value
6	OD-20220110-QNDJ41AB	RXC-81D74FBF	174995.00	High Value
7	OD-20240320-JW67SDSR	RXC-69300FE9	152991.00	High Value
8	OD-20240830-BQDMNJ9D	RXC-989B3F98	151992.00	High Value
9	OD-20231214-YTC5JADV	RXC-E63506AF	148745.75	High Value
10	OD-20240912-4MWM1QEE	RXC-7F36028E	139996.00	High Value
11	OD-20230323-QEH2CX4X	RXC-2844641C	139996.00	High Value
12	OD-20230108-XAK3JZJ2	RXC-0A8CC277	139996.00	High Value
13	OD-20231128-848P2X44	RXC-BD05B6E3	136792.80	High Value
14	OD-20221102-A7CZ6EGY	RXC-1543F5ED	135992.00	High Value
15	OD-20240412-CZNPQA4K	RXC-CD01791A	129192.40	High Value
16	OD-20220116-FJAM967S	RXC-57D736C6	125996.40	High Value

Successfully run. Total query runtime: 185 msec. 8000 rows affected.

Total rows: 8000 | Query complete 00:00:00.185 | CRLF | Ln 64, Col 28

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```
66  -- Find products that have never been ordered (use LEFT JOIN + NULL check or NOT IN subquery).
67  SELECT p.product_id AS "Product ID",
68      p.product_name AS "Product Name",
69      p.category AS "Category"
70  FROM products p
71  LEFT JOIN orders o ON p.product_id = o.product_id
72  WHERE o.order_id IS NULL;
```

Data Output | Messages | Notifications

SQL

Product ID character varying (20)	Product Name character varying (30)	Category character varying (30)

Successfully run. Total query runtime: 120 msec. 0 rows affected.

Total rows: 0 | Query complete 00:00:00.120 | CRLF | Ln 72, Col 26

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Use a window function to rank customers by total spend within each loyalty_tier. Show RANK() and DENSE_RANK().
WITH customer_spend AS (
    SELECT c.customer_id, c.customer_name, c.loyalty_tier,
        SUM(o.total_amount) AS total_spend
    FROM orders o
    JOIN customers c ON o.customer_id = c.customer_id
    GROUP BY c.customer_id, c.customer_name, c.loyalty_tier)
SELECT customer_id AS "Customer ID",
    customer_name AS "Customer Name",
    loyalty_tier AS "Loyalty Tier",
    ROUND(total_spend, 2) AS "Total Spend (₹)",
    RANK() OVER(
        PARTITION BY loyalty_tier
        ORDER BY total_spend DESC ) AS "Spend Rank",
    DENSE_RANK() OVER(
        PARTITION BY loyalty_tier
        ORDER BY total_spend DESC
    ) AS "Spend Dense Rank"
FROM customer_spend
ORDER BY "Loyalty Tier", "Spend Dense Rank" ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 2

	Customer ID character varying (20)	Customer Name character varying (100)	Loyalty Tier character varying (20)	Total Spend (₹) numeric	Spend Rank bigint	Spend Dense Rank bigint
1	RXC-64AE43C9	Naveen Dubey	Bronze	336932.25	1	1
2	RXC-5E5BC156	Divya Kumar	Bronze	263919.50	2	2
3	RXC-1405A59F	Aarav Malhotra	Bronze	160378.75	3	3
4	RXC-0A8CC277	Nisha Tiwari	Bronze	152549.00	4	4
5	RXC-57D736C6	Sneha Joshi	Bronze	130639.80	5	5
6	RXC-AC89F7D9	Priyanka Chopra	Bronze	129756.65	6	6
7	RXC-FD352E43	Rohit Agarwal	Bronze	129316.30	7	7

Successfully run. Total query runtime: 167 msec. 1964 rows affected.

Total rows: 1964 | Query complete 00:00:00.167 | CRLF | Ln 91, Col 28

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```
-- Calculate running total of revenue ordered by order_date using SUM() OVER (ORDER BY ...).
SELECT order_id AS "Order ID",
    order_date AS "Order Date",
    total_amount AS "Order Revenue (₹)",
    SUM(total_amount) OVER (
        ORDER BY order_date ASC, order_id ASC ) AS "Running Total Revenue (₹)"
FROM orders
ORDER BY order_date ASC, order_id ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 8

	Order ID character varying (30)	Order Date date	Order Revenue (₹) numeric (12,2)	Running Total Revenue (₹) numeric
1	OD-20220101-110MB0...	2022-01-01	10445.25	10445.25
2	OD-20220101-FSRJ49Y6	2022-01-01	18898.20	29343.45
3	OD-20220101-PZWWY...	2022-01-01	1998.00	31341.45
4	OD-20220101-ZKRNR5...	2022-01-01	4491.00	35832.45
5	OD-20220102-3EQHQX...	2022-01-02	2547.00	38379.45
6	OD-20220102-6MTA3S...	2022-01-02	2847.15	41226.60
7	OD-20220102-7TRR39SR	2022-01-02	3324.05	44550.65
8	OD-20220102-9JZ0C0NK	2022-01-02	823.50	45374.15
9	OD-20220102-JPWW1...	2022-01-02	5215.50	50589.65
10	OD-20220102-QJHEXG...	2022-01-02	7124.25	57713.90
11	OD-20220102-TPJN8H20	2022-01-02	12599.10	70313.00
12	OD-20220102-XVB0GE...	2022-01-02	189.05	70502.05
13	OD-20220103-7N4Z6A...	2022-01-03	1424.05	71926.10
14	OD-20220103-FQ4HT0...	2022-01-03	848.30	72774.40
15	OD-20220103-MVQJES...	2022-01-03	7196.40	79970.80
16	OD-20220103-V7CNDK...	2022-01-03	17495.00	97465.80

Successfully run. Total query runtime: 183 msec. 8000 rows affected.

Total rows: 8000 | Query complete 00:00:00.183 | CRLF | Ln 102, Col 39

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Find the percentage contribution of each category to total revenue using a subquery or CTE.
WITH category_revenue AS (
    SELECT p.category AS product_category,
        SUM(o.total_amount) AS category_total
    FROM orders o
    JOIN products p ON o.product_id = p.product_id
    GROUP BY p.category
)
SELECT product_category AS "Product Category",
    ROUND(category_total, 2) AS "Category Revenue (₹)",
    ROUND((category_total / (SELECT SUM(category_total) FROM category_revenue)) * 100, 2) AS "Percentage Contribution (%)"
FROM category_revenue
ORDER BY "Percentage Contribution (%)" DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10 | Page No: 1 of 1

	Product Category character varying (30)	Category Revenue (₹) numeric	Percentage Contribution (%) numeric
1	Furniture	18815321.35	41.21
2	Electronics	7603212.00	16.65
3	Home & Kitchen	6234356.65	13.65
4	Toys	2794009.10	6.12
5	Clothing	2753357.75	6.03
6	Sports	2301295.20	5.04
7	Beauty	1986212.25	4.35
8	Grocery	1270876.10	2.78
9	Books	1130292.15	2.48
10	Stationery	768101.70	1.68

Successfully run. Total query runtime: 120 msec. 10 rows affected.

Total rows: 10 | Query complete 00:00:00.120 | CRLF | Ln 116, Col 45

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- For each customer, find the days between their first and last order (customer lifetime in days).
SELECT o.customer_id AS "Customer ID",
    c.customer_name AS "Customer Name",
    c.loyalty_tier AS "Loyalty Tier",
    MIN(o.order_date) AS "First Order Date",
    MAX(o.order_date) AS "Last Order Date",
    (MAX(o.order_date) - MIN(o.order_date)) AS "Customer Lifetime (Days)"
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY o.customer_id, c.customer_name, c.loyalty_tier
ORDER BY "Customer Lifetime (Days)" DESC, "Customer Name" ASC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000 | Page No: 1 of 2

	Customer ID character varying (20)	Customer Name character varying (100)	Loyalty Tier character varying (20)	First Order Date date	Last Order Date date	Customer Lifetime (Days) integer
1	RXC-644CDB5B	Tarun Rao	Bronze	2022-01-05	2024-12-28	1088
2	RXC-3279434E	Gaurav Iyer	Gold	2022-01-05	2024-12-23	1083
3	RXC-F37E8199	Deepika Bajaj	Bronze	2022-01-13	2024-12-29	1081
4	RXC-C1001F2A	Naveen Tiwari	Bronze	2022-01-06	2024-12-22	1081
5	RXC-01805354	Divya Reddy	Gold	2022-01-05	2024-12-19	1079
6	RXC-EFB277BE	Sachin Bajaj	Silver	2022-01-07	2024-12-21	1079
7	RXC-E8AC913F	Suresh Iyer	Silver	2022-01-03	2024-12-16	1078
8	RXC-7BF08004	Sachin Joshi	Silver	2022-01-09	2024-12-20	1076
9	RXC-597F4A2F	Sneha Joshi	Platinum	2022-01-19	2024-12-28	1074
10	RXC-738B6B94	Sneha Singh	Platinum	2022-01-25	2024-12-27	1067
11	RXC-DEE2A74F	Shreya Kapoor	Bronze	2022-01-09	2024-12-10	1066
12	RXC-AF87407F	Deepa Malhotra	Bronze	2022-01-17	2024-12-17	1065
13	RXC-B4D9F465	Simran Bajaj	Silver	2022-01-17	2024-12-15	1063
14	RXC-6A2148B6	Deepika Rao	Platinum	2022-01-10	2024-12-06	106

Successfully run. Total query runtime: 98 msec. 1964 rows affected.

Total rows: 1964 | Query complete 00:00:00.098 | CRLF | Ln 128, Col 63

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```
129
130 -- Create a VIEW called v_order_summary that shows order_id, customer_name, product_name, category, total_amount, loyalty_tier.
131 CREATE OR REPLACE VIEW v_order_summary AS
132 SELECT o.order_id, c.customer_name, p.product_name, p.category, o.total_amount, c.loyalty_tier
133 FROM orders o
134 JOIN customers c ON o.customer_id = c.customer_id
135 JOIN products p ON o.product_id = p.product_id;
136
137 SELECT * FROM v_order_summary
138 ORDER BY total_amount DESC
139 LIMIT 5;
```

Data Output | Messages | Notifications

SQL

Showing rows: 1 to 5 | Page No: 1 of 1

	order_id character varying (30)	customer_name character varying (100)	product_name character varying (30)	category character varying (30)	total_amount numeric (12,2)	loyalty_tier character varying (20)
1	OD-20220821-VFXPPS...	Naveen Dubey	Sofa 3-seater	Furniture	314991.00	Bronze
2	OD-20220424-800MRH...	Vikram Agarwal	Sofa 3-seater	Furniture	251992.80	Silver
3	OD-20221106-1ZS8XD...	Divya Kumar	Sofa 3-seater	Furniture	251992.80	Bronze
4	OD-20220716-K7VQST...	Amit Tiwari	Sofa 3-seater	Furniture	209994.00	Silver
5	OD-20221125-XPDV9D...	Suresh Gupta	Sofa 3-seater	Furniture	209994.00	Platinum

Total rows: 5 | Query complete 00:00:00.117 | CRLF | Ln 139, Col 9

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```sql
-- Identify and count NULL values in each column of the orders table. Write a single query.
SELECT
    SUM(CASE WHEN order_id IS NULL THEN 1 ELSE 0 END) AS order_id_nulls,
    SUM(CASE WHEN customer_id IS NULL THEN 1 ELSE 0 END) AS customer_id_nulls,
    SUM(CASE WHEN product_id IS NULL THEN 1 ELSE 0 END) AS product_id_nulls,
    SUM(CASE WHEN order_date IS NULL THEN 1 ELSE 0 END) AS order_date_nulls,
    SUM(CASE WHEN quantity IS NULL THEN 1 ELSE 0 END) AS quantity_nulls,
    SUM(CASE WHEN unit_price IS NULL THEN 1 ELSE 0 END) AS unit_price_nulls,
    SUM(CASE WHEN discount_pct IS NULL THEN 1 ELSE 0 END) AS discount_pct_nulls,
    SUM(CASE WHEN discount_amount IS NULL THEN 1 ELSE 0 END) AS discount_amount_nulls,
    SUM(CASE WHEN total_amount IS NULL THEN 1 ELSE 0 END) AS total_amount_nulls,
    SUM(CASE WHEN payment_method IS NULL THEN 1 ELSE 0 END) AS payment_method_nulls,
    SUM(CASE WHEN order_status IS NULL THEN 1 ELSE 0 END) AS order_status_nulls,
    SUM(CASE WHEN delivery_days IS NULL THEN 1 ELSE 0 END) AS delivery_days_nulls,
    SUM(CASE WHEN region IS NULL THEN 1 ELSE 0 END) AS region_nulls
FROM orders;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	order_id_nulls bigint	customer_id_nulls bigint	product_id_nulls bigint	order_date_nulls bigint	quantity_nulls bigint	unit_price_nulls bigint	discount_pct_nulls bigint	discount_amount_nulls bigint	total_amount_nulls bigint	payment_method_nulls bigint	order_status_n bigint
1	0	0	0	0	0	0	0	0	0	0	

Successfully run. Total query runtime: 84 msec. 1 rows affected.

Total rows: 1 | Query complete 00:00:00.084 | CRLF | Ln 156, Col 13

Capstone/postgres@PostgreSQL 18

No limit

Query | Query History

```
    SUM(CASE WHEN order_status IS NULL THEN 1 ELSE 0 END) AS order_status_nulls,
    SUM(CASE WHEN delivery_days IS NULL THEN 1 ELSE 0 END) AS delivery_days_nulls,
    SUM(CASE WHEN region IS NULL THEN 1 ELSE 0 END) AS region_nulls
FROM orders;

-- For the marketing table: calculate CTR (clicks/impressions × 100), Conversion Rate (conversions/clicks × 100), and ROAS (revenue/spend). Round to 2 decim
SELECT campaign_id AS "Campaign ID",
    campaign_name AS "Campaign Name",
    channel AS "Channel",
    spend_inr AS "Ad Spend (₹)",
    revenue_generated AS "Revenue Generated (₹)",
    ROUND((clicks::NUMERIC / NULLIF(impressions, 0)) * 100, 2) AS "CTR (%)",
    ROUND((conversions::NUMERIC / NULLIF(clicks, 0)) * 100, 2) AS "Conversion Rate (%)",
    ROUND((revenue_generated::NUMERIC / NULLIF(spend_inr, 0)), 2) AS "ROAS"
FROM marketing
ORDER BY "ROAS" DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 200 | Page No: 1 of 1

	Campaign ID character varying (20)	Campaign Name character varying (30)	Channel character varying (30)	Ad Spend (₹) numeric	Revenue Generated (₹) numeric	CTR (%) numeric	Conversion Rate (%) numeric	ROAS numeric
1	CAMP043	Mega Weekend	Google Ads	6799	10491086.0	6.34	5.25	1543.03
2	CAMP108	Festive Bonanza	Google Ads	6321	9446811.0	5.08	7.60	1494.51
3	CAMP117	Flash Friday	Instagram	9179	12815653.0	3.97	9.70	1396.19
4	CAMP036	Back to School	Push Notification	17551	18512385.0	6.44	14.45	1054.78
5	CAMP048	Referral Drive	YouTube	32754	19269315.0	6.88	10.09	588.30
6	CAMP168	Referral Drive	Influencer	10325	6014998.0	4.05	14.38	582.57
7	CAMP029	Loyalty Rewards	YouTube	5850	2098244.0	1.63	8.17	358.67
8	CAMP040	End of Season	Email	23735	8072892.0	7.99	5.26	340.13
9	CAMP162	Clearance Sale	Email	74288	22881215.0	6.57		

Successfully run. Total query runtime: 112 msec. 200 rows affected.

Total rows: 200 | Query complete 00:00:00.112 | CRLF | Ln 168, Col 22